



CONCUSO NACIONAL DE QUÍMICA 2019

"Imagination is more important than knowledge."

Albert Einstein

Name(s): _____ Last name(s): _____
School: _____
Municipality: _____ Province: _____
CODE: _____ Test: **DAY 2. TWELFTH GRADE EXAM**

Instructions

- Write clearly all your data in the appropriate place (names, last names, school, municipality, province). Do not write anything in the field CODE.
- Verify that your questions booklet is complete. This is the Day 2 problem set, with an individual exam for 12th grade. The exam has 3 problems (problem 9, 7 items; problem 10, 4 items; problem 11, 10 items) and a total of 6 pages.
- All data, constants, conversions and basic equations needed are included in the exam. You are allowed to use a scientific calculator.
- At the beginning of each problem its scoring (from a total of 100 points), each item scoring (in marks) is indicated.

Success!



Ministerio de Educación
República de Cuba



CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Equation of ideal gas	$p \cdot V = n \cdot R \cdot T$
Gases constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta E + W$
Zero on Celsius scale	273,15 K	Gibbs' free energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$	$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

PERIODIC TABLE OF ELEMENTS

1 1A																		18 8A	
1 H 1.008	2 2A																	2 He 4.003	
3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18		
11 Na 22.99	12 Mg 24.31	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95		
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80		
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3		
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)		
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Ds (269)	111 Rg (272)	112 Uub (277)		114 Uuq (277)		116 Uuh (277)		118 Uuo (277)		
58 Ce 140.1		59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0					
90 Th 232.0		91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)					

Problem 9. Electrochemistry (15 points)

9.1	9.2	9.3	9.4	9.5	9.6	9.7	Total	Total
7	3	4	4	3	3	6	30	15

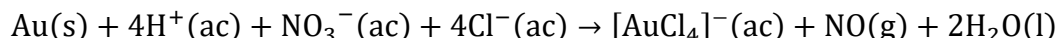
Copper, silver and gold are transition elements belonging to the 11th group in the periodic table. Their corresponding simple substances are metals and are employed since the antiquity for the manufacture of coins.

Highly pure copper (higher than 99,99%) is obtained by electrodeposition in a copper cathode. In the electrolytic cell impure copper is placed in the anode and in the cathode a plate made of pure copper is placed. An aqueous solution of copper(II) sulfate is employed as electrolytic conductor.

9.1. Suppose that in the electrolytic cell described above a copper anode containing impurities of gold, platinum and silver is placed. Write all the possible anodic and cathodic half reactions and select which ones are thermodynamically more favored. Assume standard conditions.

9.2. A copper anode has approximate dimensions of 100×125 cm, a thick of 5 cm and an approximate weight of 350 kg. Calculate the minimum amount of current (in Coulomb) to carry out the electrorefining of a copper anode with an approximate purity of 96%.

Copper, silver and gold are not susceptible to oxidation by H⁺ ions in standard conditions and therefore they are employed, together with platinum, in ornaments and jewelry. Nitric acid is able to dissolve copper and silver; while aqua regia, a 3:1 mixture of concentrated hydrochloric and nitric acids, is an effective reagent for the oxidation of gold and platinum. In aqua regia, NO₃⁻ ions provide the oxidizing power and Cl⁻ ions act as complexing agents. The total reaction occurring with gold is:



9.3. Demonstrate that the oxidation of copper and silver with nitric acid is spontaneous.

9.4. Write half-equations for the oxidation and reduction corresponding to the dissolution of gold with aqua regia.

It is believed that the active oxidizing species in the aqua regia solution are chlorine and nitrosyl chloride, generated by the reaction



9.5. Represent the structure of NOCl molecule.

Gold is extracted by lixiviation with cyanide. Using facilitates gold oxidation through the formation of complex ion [Au(CN)₂]⁻. To recover the dissolved gold a reductant reagent is employed, for example, zinc dust. Today exist several communities all around the globe that fight against the negative impacts in economy, society and environment caused by the mining activity for obtaining gold using cyanide.

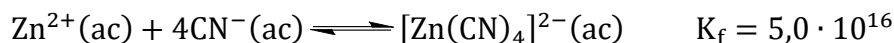
9.6. Write the chemical equation corresponding to the reduction of gold complex ion with zinc.

9.7. Calculate the constant of formation K_f of complex ion [Au(CN)₂]⁻ from ions Au⁺ and CN⁻ with the following data.



$E^\circ(\text{V})$ para $\text{pH} = 0$:

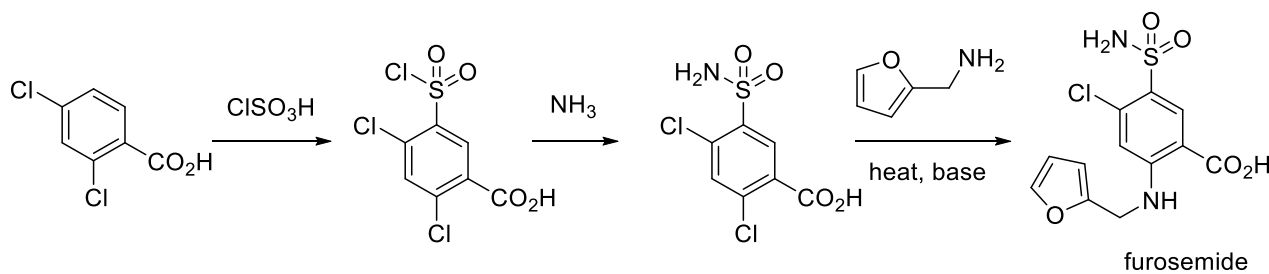
$\text{Cu}^{2+}(\text{ac})/\text{Cu}(\text{s}) = 0,340$; $\text{Au}^{3+}(\text{ac})/\text{Au}(\text{s}) = 1,52$; $\text{Pt}^{2+}(\text{ac})/\text{Pt}(\text{s}) = 1,188$; $\text{Ag}^{+}(\text{ac})/\text{Ag}(\text{s}) = 0,799$;
 $\text{O}_2(\text{g})/\text{H}_2\text{O}(\text{l}) = 1,229$; $\text{H}^{+}(\text{ac})/\text{H}_2(\text{g}) = 0,00$; $\text{S}_2\text{O}_8^{2-}(\text{ac})/\text{SO}_4^{2-}(\text{ac}) = 1,96$; $\text{SO}_4^{2-}(\text{ac})/\text{SO}_2(\text{g}) = 0,158$;
 $\text{NO}_3^{-}(\text{ac})/\text{NO}(\text{g}) = 0,96$; $[\text{Au}(\text{CN})_2]^{-}(\text{ac})/\text{Au}(\text{s}) = -0,596$; $\text{Au}^{+}(\text{ac})/\text{Au}(\text{s}) = 1,83$.



Problem 10. Organic Chemistry I (15 points)

10.1	10.2	10.3	10.4	Total	Total
6	10	8	6	30	15

Diuretics are substances that induce the release of water and electrolytes, reducing the volume of liquids inside the organism. Furosemide is an exceptionally potent diuretic used for the treatment of edemas and hypertension. The preparation of this drug is shown.



10.1. Propose a synthetic rout to obtain 2,4-dichlorobenzoic acid starting from toluene.

10.2. Suggest a mechanism for the chlorosulfonation reaction. Represent all the possible resonance structures of the intermediate formed.

10.3. Suggest a mechanism for the reaction with furfurylamine. Represent all the possible resonance structures of the intermediate formed.

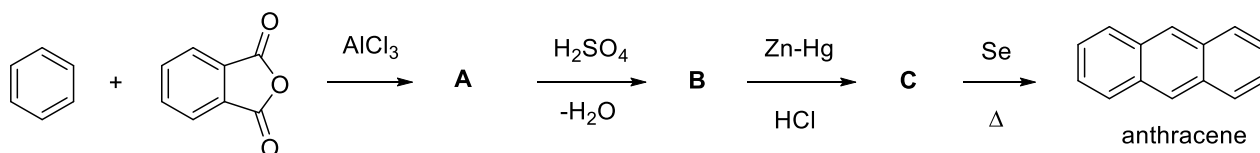
10.4. In the synthetic sequence both nucleophilic and electrophilic aromatic substitution reactions are employed. Compare both types of reactions briefly according to: substituent effect, regioselectivity, and charge of the intermediate formed.

Problem 11. Organic Chemistry II (20 points)

11.1	11.2	11.3	11.4	11.5	11.6	11.7	11.8	11.9	11.10	Total	Total
6	3	6	2	2	3	3	3	6	6	40	20

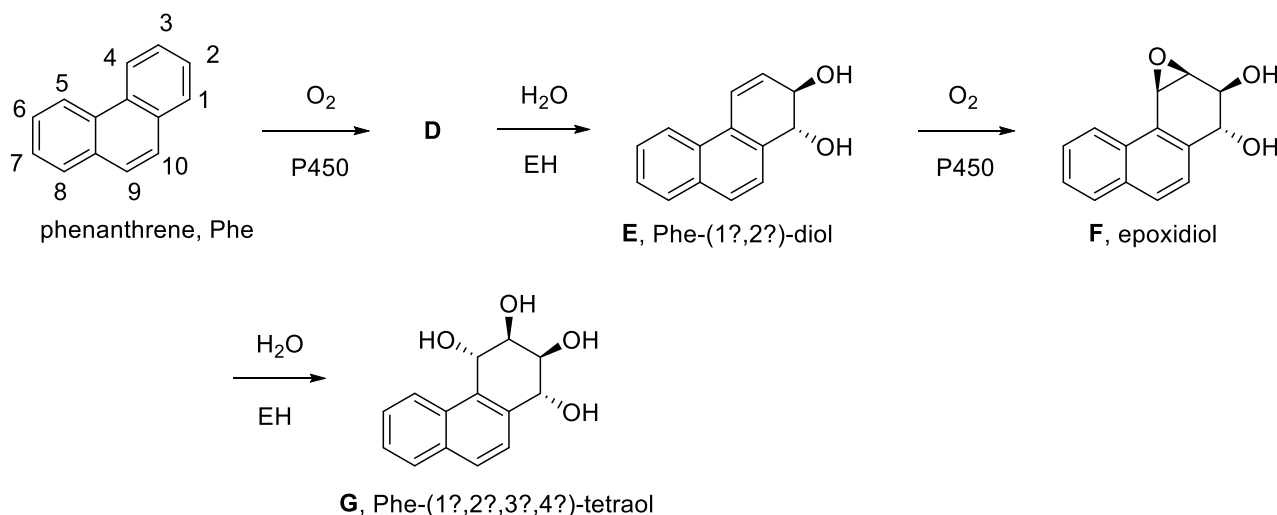
Polycyclic aromatic hydrocarbon (PAH) are strong contaminants found in oil, coal and tar deposits. These compounds are also formed as by-products of fuel combustion.

Bellow the Haworth's synthesis of anthracene (R. D. Haworth, J. Chem. Soc. 1932, 1125) is sketch, starting from benzene and phthalic anhydride.



11.1. Identify compounds **A-C** in the synthetic route towards anthracene.

The degradation of PAH inside the human being comprises a series of reactions of oxidation, substitution and hydrolysis aiming to their transformation into soluble substances that can be then easily excreted. Enzyme complex cytochrome P450 participates in the processes to oxidize PAH. The most common reaction catalyzed by P450 is the insertion of an oxygen atom in an organic substrate. Some of the compounds derived from the action of P450 on PAH are very reactive epoxides which can react with nitrogen bases present in the desoxyribonucleic acid (DNA). This is the cause for numerous PAH having strong carcinogenic, teratogenic and mutagenic effects. On the other hand, intermediates can add water, a reaction that is catalyzed by the enzyme epoxide hydrolase (EH). A portion of the metabolic route for the degradation of phenanthrene is shown.



11.2. Propose a structure for compound **D** in the phenanthrene route.

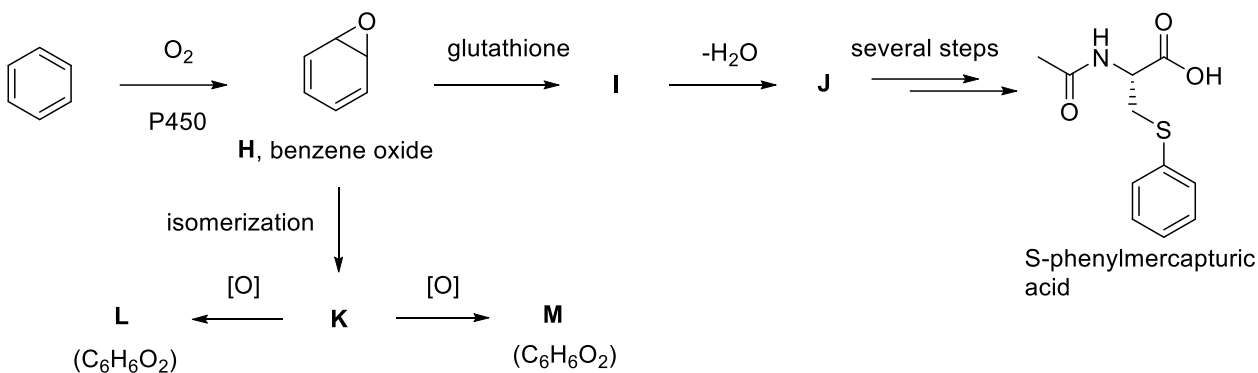
11.3. Suggest a mechanism for the hydrolysis of **F** forming **G** in acid media.

11.4. Assign the absolute configuration of all the asymmetric carbons of tetraol **G**.

11.5. The enzymatic action on phenanthrene also originates the enantiomer of **G** as product. Represent its structure.

11.6. Epoxidiol **F** can react with nitrogen bases present in DNA. Suggest the structure of the product of the reaction between **F** and a fragment of DNA containing guanine (see the figure at the end of the problem).

Benzene and toluene are degraded in the organism in a similar fashion to PAH. Fragments of their metabolic corresponding routes appear underneath.



11.7. Benzene oxidation by P450 originates highly reactive compound **H**, which exists in the form of two isomers in equilibrium, none of them is aromatic. Propose a structure for the isomers of **H** non represented in the scheme.

Glutathione (see the structure below) is a tripeptide existing in the organism that helps to protect the cells from reactive species such as free radicals, peroxide, epoxides, epoxides and Michel acceptors. The thiol group present is the responsible of the biological activity of glutathione and can function as proton donor, nucleophile and reductant.

11.8. Represent the structure of the product of the reaction between ethyl acrylate ($\text{H}_2\text{C}=\text{CHCO}_2\text{Et}$) and glutathione.

11.9. The reaction between benzene oxide and glutathione originates compound **I**, which dehydrates quickly to form **J**. Propose structures for compounds **I** and **J**.

Isomerization of benzene oxide forms **K**, a well-known compound whose subsequent oxidation originates un isomers **L** and **M**. All these compounds are aromatic and react with sodium hydroxide. **L** and **M** are easily oxidized by air to form colored substances.

11.10. Represent the structures of compounds **K-M**.

